

Feature-Substrate Spec

Per-page memory-delta metrics — the families A–G (+H)

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This spec catalogues the per-page memory-delta metrics for the capture feature-substrate -- the set of primitives we extract once, in-pass, from each changed 4 KB page before its snapshot is deleted, so that all later feature engineering is offline with no recapture.

The generative idea: a metric is a choice on four axes -- representation x reading x reference x unit. Hamming = bit-space x amount x consecutive x page. Cosine = vector-space x direction x consecutive x page. Varying any axis generates the families below. A = amount (vary the space), B = direction (vary the reading), C = content of q, D = internal structure, E = the field as an image, F = the field over time, G = the reference, H = learned representations.

Notation: p = old page, q = new page (4096 bytes each); P, Q = their 256-bin byte histograms as probabilities; C() = a compressor. Each family is one perspective -- a per-unit heat-map channel.

How to read this

Each family is one perspective (a per-unit heat-map channel). Every family has a full table: the metric, what it **Measures** (formula / definition), and **What it catches** that the others miss. Family A is split by representation **Space**; Family B by **Reading**; the rest by their natural groups.

Phasing: Phase 1 (A–F) vs Phase 2 (G, H)

Phase 1 -- Families A-F. All fit the existing capture FORM: a closed-form function of the local consecutive diff (prev vs curr), computed in the same page-comparison pass the APF differ already runs, stored as a small per-page descriptor. This is a small, additive, flag-gated emit (mirroring the Plan 06 disk block) -- byte-identical when off. It does NOT require redesigning or re-proving the capture architecture; it is a small change within the proven form.

Phase 2 -- Families G and H (deferred). These break the form. G needs a new reference / state model (pin a baseline snapshot, keep an EWMA accumulator, or buffer N lagged snapshots -- the pipeline deletes prev immediately today). H's learned half needs a trained model and a two-pass capture (store page samples -> train -> run inference at capture) plus a model registry. Both may require redesigning the capture, so they are research extensions, not the first substrate.

Family A — Amount of change (alternatives to Hamming)

Amount = how much one page changed. Hamming measures it in bit-position space. The SAME question has different answers in three representation SPACES: positional (where the bytes sit), distributional (the byte histogram, position-free), and informational (compressibility). A memmove is huge positionally but ~zero distributionally; encryption is huge distributionally and informationally. Each row is a magnitude heat-map channel.

Space	Metric	Measures	What it catches (vs Hamming)
Positional	Hamming (bit)	popcount(p XOR q)	bits flipped -- total bit churn (the baseline)
Positional	Byte-Hamming / L0	count(p_j != q_j)	how many BYTES changed (vs how many bits) -- spread of change
Positional	L1 / SAD	sum abs(p_j - q_j)	total byte-magnitude moved -- weights big byte jumps, not just that a byte changed
Positional	L2 / Euclidean	sqrt(sum (p_j - q_j)^2)	Euclidean magnitude -- emphasises a few large byte changes

Space	Metric	Measures	What it catches (vs Hamming)
Positional	L-inf / Chebyshev	$\max \text{abs}(p_j - q_j)$	the single biggest byte jump -- an outlier change a page-wide sum hides
Positional	Mean abs change	$\text{mean abs}(p_j - q_j)$	average per-byte change -- size-normalised magnitude
Positional	Gradient-magnitude change	$\text{energy of } \text{abs}(\text{grad } q) - \text{abs}(\text{grad } p)$	change in local roughness -- did the page get smoother/jaggier
Distributional	Total Variation	$0.5 * \sum \text{abs}(P_b - Q_b)$	max probability mass that moved -- coarse histogram shift
Distributional	Chi-square	$\sum (P_b - Q_b)^2 / (P_b + Q_b)$	frequency-weighted histogram diff -- sensitive to spikes / rare bins
Distributional	Hellinger	$\sqrt{\sum (\sqrt{P_b} - \sqrt{Q_b})^2} / \sqrt{2}$	bounded [0,1] metric -- balanced, well-behaved histogram distance
Distributional	Kullback-Leibler	$\sum P_b * \log(P_b / Q_b)$	info lost approximating one by the other -- raw surprise (asymmetric, unbounded)
Distributional	Jensen-Shannon	symmetrised, bounded KL	symmetric bounded version -- survives non-overlapping byte supports
Distributional	Wasserstein / EMD	min transport cost $P \rightarrow Q$	how FAR byte-mass moved (geometry-aware) -- gradual shift vs spike; order matters
Distributional	Bhattacharyya	$-\ln \sum \sqrt{P_b * Q_b}$	distribution overlap -- close cousin of Hellinger
Distributional	Histogram intersection	$\sum \min(P_b, Q_b)$	shared mass -- simple overlap similarity
Informational	Entropy delta	$H(q) - H(p)$	change in randomness -- toward/away from ciphertext (signed = also a direction)
Informational	Normalized Compression Distance	$(C(pq) - \min) / \max$	shared compressible structure -- weak on high-entropy (encrypted) pages
Informational	Compressed-size delta	$\text{len}(C(q)) - \text{len}(C(p))$	change in raw complexity -- page got simpler / more complex
Informational	Structural-entropy change	change in entropy-stream / wavelet signature	byte-granularity structural shift -- the malware-classification feature
Informational	LZ-complexity change	change in LZ77 factor count	algorithmic-complexity change -- Kolmogorov proxy

Family B — Direction of change (alternatives to Cosine)

Direction = which WAY a page changed. Cosine reads only the STRUCTURE (pattern) component. SSIM shows that similarity decomposes into structure + level + spread; add polarity, distributional-direction, and spatial-shift. Each row's direction sense says which way the change points.

Reading	Metric	Measures	Direction sense (what it catches)
Structure	Cosine similarity	$\text{dot}(p, q) / (\text{norm}(p) * \text{norm}(q))$	angle between content vectors -- did the pattern stay aligned (the baseline)
Structure	Pearson correlation	centered cosine (mean removed)	linear co-variation ignoring level shift -- pattern alignment minus brightness
Structure	Spearman rank corr.	correlation of byte ranks	did the ORDERING of byte values persist -- monotonic structure even if rescaled
Structure	Kendall tau	concordant-pair fraction	ordinal agreement -- robust ordering persistence
Structure	SSIM structure term	$\text{cov}(p, q) / (\text{std}(p) * \text{std}(q))$	local structural similarity -- the 's' term of SSIM
Level	Mean shift (signed)	$\text{mean}(q) - \text{mean}(p)$	did bytes get bigger or smaller on average -- net 'brightness' direction
Level	SSIM luminance term	$(2 * \mu_p * \mu_q + c) / (\mu_p^2 + \mu_q^2 + c)$	closeness of means -- level agreement
Level	Median shift	$\text{median}(q) - \text{median}(p)$	robust level direction -- central-tendency move resistant to outliers

Reading	Metric	Measures	Direction sense (what it catches)
Spread	Variance ratio	$\text{var}(q) / \text{var}(p)$	got more or less varied -- densifying vs sparsifying direction
Spread	Std-dev delta (signed)	$\text{std}(q) - \text{std}(p)$	change in spread -- additive form of the contrast direction
Spread	SSIM contrast term	$(2 * \text{std}_p * \text{std}_q + c) / (\text{var}_p + \text{var}_q + c)$	spread agreement -- the 'c' term of SSIM
Spread	Range / IQR delta	change in (max-min) or IQR	dynamic-range direction -- widened vs narrowed value range
Polarity	Fraction up vs down	$\text{count}(q_j > p_j)$ vs $\text{count}(q_j < p_j)$	which way the changed bytes moved -- the directional 'current'
Polarity	Net signed drift	$\text{sum}(q_j - p_j)$	net byte-value flow -- accumulation vs erasure
Polarity	Sign-of-delta entropy	entropy of the up/down pattern	is the direction coherent or a mixed-up wash
Distributional dir.	Entropy delta sign	$\text{sign}(H(q) - H(p))$	toward random or toward structured -- encryption vs zeroing direction
Distributional dir.	Histogram-mean shift sign	sign of byte-value mean move	toward high or low byte values
Distributional dir.	Move-toward-uniform	$KL(q \text{unif}) - KL(p \text{unif})$	approaching ciphertext -- uniformisation direction
Distributional dir.	Zero-mass delta	$Q_0 - P_0$ (zero-bin mass change)	toward zeroed / freed memory
Spatial-shift	Cross-correlation lag	$\text{argmax circular } \text{xcorr}(p, q)$	content SHIFTED within the page (memmove/scroll) -- offset & direction
Spatial-shift	Phase-correlation peak	peak of normalised cross-power spectrum (FFT)	translational shift -- robust shift estimate
Spatial-shift	Optimal byte-rotation	best cyclic alignment offset	content rotated within the page

Family C — Content / character of q (what it became; reference-free)

Not amount, not direction -- describe the NEW page q itself, with no reference to p. Reference-free fingerprints of what the page became (ciphertext, zeros, text, pointers, ...).

Metric	Measures	What it catches
Shannon entropy of q	$-\text{sum } P_b * \log_2(P_b)$	randomness / compressibility -- ciphertext (~8) vs structured (4-5) vs sparse (low)
Structural entropy	entropy stream / wavelet energy over the page	texture of randomness -- localised encrypted regions
Distinct byte count	$\text{count}(b: \text{count}_b > 0)$	palette size -- full 256 = random/dense; few = structured
Zero-byte fraction	$\text{count}(\text{byte} == 0) / 4096$	emptiness -- zeroed / freed / sparse pages
Constant-fill fraction	$\text{max}_b \text{count}(b) / 4096$ (e.g. 0xFF)	fill patterns -- uninitialised / poisoned memory
Printable-ASCII fraction	$\text{count}(0x20..0x7E) / 4096$	text-likeness -- strings / JSON / code vs binary
Byte moments	mean, var, skew, kurtosis of bytes	distribution shape -- coarse content fingerprint
Bigram entropy	entropy over adjacent byte pairs	local structure -- pointers / tables vs random
Autocorrelation peak	max self-correlation at lag > 0	periodicity within the page -- arrays / records / strides
Chi-square uniformity	$\text{sum}(\text{count}_b - E)^2 / E$	deviation from uniform -- the 'is it random' (encryption) test

Family D — Internal structure / texture (where & how, within the unit)

WHERE the change sits inside the page, and how textured the content is. Two groups: the intra-page change mask (location of the change), and the texture of q (its spatial pattern).

Group	Metric	Measures	What it catches
Change location	Changed-run count	number of contiguous changed-byte segments	clustered vs scattered change -- localised edit vs full rewrite
Change location	Change span	last - first changed offset	extent of the touched region -- narrow patch vs whole page
Change location	Change centroid	center of mass of changed offsets	where in the page -- header vs body edits
Change location	Longest changed run	max contiguous changed bytes	biggest edited block -- block-overwrite size
Change location	Change density in span	changed bytes / span	compactness of the edit
Texture of q	Gradient / edge energy	$\sum \text{abs}(q_{[j+1]} - q_{[j]})$	local roughness -- smooth (zeros/text) vs jagged (random/binary)
Texture of q	GLCM / Haralick	contrast, homogeneity, energy, correlation of byte co-occurrence	texture class -- structured vs noise
Texture of q	LBP histogram	local binary patterns over the byte sequence	micro-texture -- repeated motifs
Texture of q	Spatial-frequency energy	low- vs high-band FFT/DCT energy	coarse vs fine structure -- periodic tables vs random
Texture of q	Run-length distribution	lengths of constant-byte runs	fill / structure -- RLE-compressibility proxy

Family E — Spatial field across pages (the heat map AS an image)

Lay any per-page channel (amount, direction, ...) across physical address -> a 2D image (512x512 for a 1 GB guest). These describe the change FIELD as an image. All are DERIVED offline from a per-page channel + page_index -- nothing extra is stored.

Metric	Measures	What it catches
Active-area ratio (APF)	changed pages / total pages	overall breadth -- the foreground fraction (the baseline)
Moran's I	spatial autocorrelation of the field	clustered / dispersed / random layout -- sequential sweep vs scattered random
Connected-component count	number of contiguous changed-page regions	fragmentation -- one region vs many specks
Component size distribution	sizes of changed-page clusters	region scale -- big buffers vs scattered touches
Address-axis FFT / autocorr.	periodicity of the changed-page index set	stride detection -- strided access (every Nth page)
GLCM of the magnitude field	Haralick texture of the heat map	spatial change pattern / texture
Concentration (Gini / entropy)	inequality of magnitude across pages	a few hot pages vs uniform spread
Field edges / gradient	boundaries of active regions	working-set edges -- where activity starts/stops in address space
Centroid & dispersion	center and spread of the active set	WHERE in RAM the work sits, and how concentrated

Family F — Temporal dynamics (the heat-map movie; needs page identity)

Per-page over snapshots (requires storing page_index). The heat map as a MOVIE: how pages and the active region evolve over time. All DERIVED offline from page_index + the per-page channels.

Level	Metric	Measures	What it catches
Per-page	Revisit rate	snapshots a page changes / total	page hotness -- hot loop (high) vs touched-once (low)
Per-page	Inter-change interval	gaps between a page's changes	rhythm -- steady vs bursty
Per-page	Burstiness	Fano factor / CV of intervals	clumping of a page's writes
Per-page	Change-series autocorr.	periodicity of a page's activity over time	cyclic access to the same page

Level	Metric	Measures	What it catches
Per-page	Page change-point	when a page's behaviour shifts	per-page phase transitions
Per-page	Page lifetime	first-touch to last-touch span	allocation -> free lifetime
Field-level	Working-set size	distinct pages active in a window	footprint dynamics -- growing vs stable
Field-level	Working-set migration	Jaccard between consecutive active sets	does the hot region MOVE -- sweep (low overlap) vs hot loop (high)
Field-level	Active-region drift	displacement of the activity centroid	sweep direction & speed (optical-flow of the heat map)
Field-level	New-page rate	newly-touched pages per step	allocation / discovery rate

Family G — Reference basis (what we diff against; multiplies A-F)

Every metric above diffs against SOMETHING. Hamming/cosine use t vs t-1. Changing the reference is a whole perspective family -- the memory-forensics standard is diff-vs-baseline. This multiplies every metric in A-F. Chosen offline by picking the reference snapshot. PHASE 2 (deferred): the non-consecutive references need a NEW capture state model -- a pinned baseline, an EWMA accumulator, or buffered lagged snapshots (the pipeline deletes prev immediately today). Only the histogram-space references are offline-derivable; positional baseline/lag diffs need the raw bytes at both times. May require a capture redesign.

Metric	Measures	What it catches
Consecutive (t vs t-1)	adjacent snapshots	instantaneous change -- the baseline (what Hamming/cosine use)
Baseline / idle	vs a fixed idle snapshot	drift from rest -- total deviation from 'doing nothing' (forensics standard)
Self EWMA	vs the page's own moving average	self-anomaly -- unusual relative to that page's own habit
N-lag / multi-scale	vs t-k for several k	slow drift vs fast churn -- separates timescales
Boot / origin	vs the first snapshot	cumulative change since start -- lifetime accumulation
Canonical page	vs a reference page (e.g. zero page)	absolute content type -- zero-distance, fill-distance

Family H — Learned / embedding (appendix; heavier; future work)

Representations learned from data rather than computed in closed form. Heavier (training, and usually need the raw bytes), so out of scope for the first substrate, but listed for completeness as a distinct perspective family. PHASE 2 (deferred): the learned half (autoencoder, learned embeddings) needs a trained model + a TWO-PASS capture (store page samples -> train offline -> run inference at capture) + a model registry -- a new modelling paradigm. The fixed-hash sketches (SimHash, random projection) are the exception: data-independent, no training, so they already fit the Phase 1 form.

Metric	Measures	What it catches
Autoencoder reconstruction error	distance from a learned 'normal' page manifold	novelty of a page's content -- one-class page anomaly
Learned page embedding distance	distance in a trained embedding space	semantic page similarity
SimHash / fuzzy hash (ssdeep)	locality-sensitive hash distance	cheap near-duplicate detection
Random-projection sketch	L2 in a Johnson-Lindenstrauss sketch	cheap approximate L2 at low storage

Synthesis: channels, and what to store vs derive

Treat each per-page perspective as an image CHANNEL; the substrate is a sparse (changed pages x channels) tensor per snapshot. The decisive rule: store only what needs the raw bytes and cannot be derived. Families E, F,

and G are entirely DERIVED offline from stored per-page channels + page_index. The distributional half of A, the distributional direction of B, and all of C are derived offline from a stored byte histogram. So the in-pass store list is small; the offline experiment space is enormous.

The one real decision is the byte histogram (256 B/page, exact, but ~40 GB on a 100%-APF cell) vs a handful of cheap moments (~10 B, approximate). Likely answer: moments always, full histogram on quiet cells only. Compute budget (capture must keep up at 500 ms over 1 GB): cheap in-pass = Hamming, L0, L1, L-inf, mean, var, entropy, zero-frac, histogram; expensive = EMD, NCD, GLCM -- so those are never in-pass, they are derived offline from the histogram + index.

Channel	Size	Store / derive	Why
page_index	u32 (4 B)	STORE	unlocks all of E, F, G offline -- highest leverage single field
hamming (bit)	u16 (2 B)	STORE	positional amount; cannot come from a histogram
L1 / SAD	u32 (4 B)	STORE	positional amount, magnitude-weighted
cosine	u16 (2 B)	STORE	structure direction; needs the raw vectors
byte histogram of q	256 B (quiet cells) / moments ~10 B	STORE (tiered)	derives all of A-distributional, B-distributional, C
n_runs, span	2 B	STORE	intra-page structure (D, change-location)
everything in E, F, G	0 B	DERIVE	computed offline from page_index + the stored channels
GLCM / FFT texture (D)	0 B (or raw)	DERIVE / defer	needs raw bytes; defer unless a Tier-2 page sample is kept

Sources

[Histogram distance overview](#)

[Divergence metrics guide](#)

[SSIM components](#)

[SSIM derivation \(Wang et al.\)](#)

[NCD on malware](#)

[Detecting malware with information complexity](#)

[Image descriptors \(color/shape/texture\)](#)

[Intel PML working-set estimation](#)

[eBPF working-set-size estimation](#)

[Moran's I image segmentation](#)

[Image biomarker standardisation \(texture\)](#)